WORKED SOLUTION

STEP III 2024, Question 4

Pure • Tangents to a Parabola

STEP III 2024, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything depends on the formula for tan of the angle between two lines, which is just the $\tan(A-B)$ formula in disguise. Once the tangents at $x=p$ and $x=q$ are known to meet at a point (X,Y) with $p+q$ and pq given in terms of X and Y, any condition on the angle, which involves $(p-q)^2$ and pq, can be rewritten in X and Y alone. That rewritten condition is the locus.

Solution

Part (i)

Let the lines make angles α and β with the positive x-axis, so $m_1=\tan\alpha$ and $m_2=\tan\beta$ (both lines have gradients, so neither is vertical). The angle between the lines is $\alpha-\beta$ up to sign and up to adding multiples of 180°. If the acute angle between them is 45°, then $\alpha-\beta$ is $\pm45^\circ$ plus a multiple of 180°, and since tan has period 180°,

$$\tan(\alpha-\beta)=\tan(\pm45^\circ)=\pm1.$$

In particular $\cos(\alpha-\beta)\neq0$, so the compound-angle formula applies and $1+m_1m_2\neq0$:

$$\tan(\alpha-\beta)=\frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}=\frac{m_1-m_2}{1+m_1m_2}.$$

Hence $\dfrac{m_1-m_2}{1+m_1m_2}=\pm1$, as required. More generally, the acute angle θ between the lines satisfies $\tan\theta=\left|\dfrac{m_1-m_2}{1+m_1m_2}\right|$, which we use in part (iii).

Part (ii)

On C we have $y = \dfrac{x^2}{4a}$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x}{2a}$. The tangent at $(p, \frac{p^2}{4a})$ is

$$y - \frac{p^2}{4a} = \frac{p}{2a}(x - p) \quad \Longleftrightarrow \quad y = \frac{p}{2a}x - \frac{p^2}{4a},$$

and similarly the tangent at $x = q$ is $y = \dfrac{q}{2a}x - \dfrac{q^2}{4a}$. Subtracting,

$$\frac{p-q}{2a}\,x = \frac{p^2 - q^2}{4a} = \frac{(p-q)(p+q)}{4a}.$$

Since $p \neq q$ we may divide by $p - q$, giving $x = \frac{1}{2}(p+q)$, as required. Substituting into the first tangent,

$$y = \frac{p(p+q)}{4a} - \frac{p^2}{4a} = \frac{pq}{4a}.$$

So the tangents meet at

$$(X, Y) = \left(\tfrac{1}{2}(p+q),\ \frac{pq}{4a}\right), \qquad \text{so} \qquad p + q = 2X, \quad pq = 4aY. \tag{$*$}$$

Now suppose the tangents at $x = p$ and $x = q$ are at 45° to each other. They are certainly different lines, so $p \neq q$ and $(*)$ holds. Their gradients are $m_1 = \frac{p}{2a}$ and $m_2 = \frac{q}{2a}$, so

$$\frac{m_1 - m_2}{1 + m_1 m_2} = \frac{(p-q)/(2a)}{1 + pq/(4a^2)} = \frac{2a(p-q)}{4a^2 + pq}.$$

By part (i) this is ± 1; squaring removes the sign:

$$4a^2(p-q)^2 = (4a^2 + pq)^2.$$

Ask yourself How do we get rid of p and q when the condition involves $p - q$, not $p + q$?

Use $(p-q)^2 = (p+q)^2 - 4pq$, which is symmetric, so $(*)$ applies:

$$(p-q)^2 = 4X^2 - 16aY, \qquad 4a^2 + pq = 4a^2 + 4aY = 4a(a + Y).$$

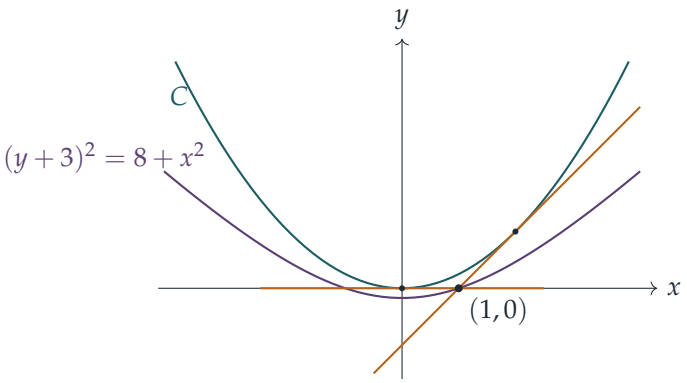
The condition becomes

$$4a^2\,(4X^2 - 16aY) = 16a^2(a + Y)^2.$$

Dividing by $16a^2$ (allowed as $a \neq 0$),

$$\begin{aligned} & X^2 - 4aY = Y^2 + 2aY + a^2, \\ \text{so} \qquad & X^2 + 8a^2 = Y^2 + 6aY + 9a^2 = (Y + 3a)^2. \end{aligned}$$

So the meeting point lies on $(y + 3a)^2 = 8a^2 + x^2$, as required.



With $a = 1$: the tangents at $x = 2$ (gradient 1) and $x = 0$ (gradient 0) are at 45°, and they meet at $(1, 0)$, which is on the upper branch of the hyperbola.

Part (iii)

Let (X, Y) be a point on $(y + 7a)^2 = 48a^2 + 3x^2$ where two tangents to C, at $x = p$ and $x = q$ with $p \neq q$, meet, so $(*)$ holds. By the remark at the end of part (i), the acute angle θ between them satisfies

$$\tan^2\theta = \left(\frac{2a(p-q)}{4a^2+pq}\right)^2 = \frac{4a^2(4X^2-16aY)}{16a^2(a+Y)^2} = \frac{X^2-4aY}{(Y+a)^2},$$

using the same substitutions as in part (ii). (If $Y = -a$ the tangents would be perpendicular; we show below that this cannot happen on this curve.)

? **Ask yourself** How does the equation of the curve help with $X^2 - 4aY$?

Expand the curve's equation: $Y^2 + 14aY + 49a^2 = 48a^2 + 3X^2$, that is,

$$3X^2 = Y^2 + 14aY + a^2.$$

Then

$$3(X^2 - 4aY) = Y^2 + 14aY + a^2 - 12aY = Y^2 + 2aY + a^2 = (Y + a)^2.$$

Two consequences follow. First, $Y \neq -a$: otherwise $X^2 - 4aY = 0$, i.e. $X^2 = -4a^2$, impossible as $a \neq 0$. So $X^2 - 4aY = \frac{1}{3}(Y + a)^2 > 0$, which also confirms that the point lies outside C: the quadratic $t^2 - 2Xt + 4aY = 0$, whose roots are p and q, has two distinct real roots, so there really are two tangents through every point of the curve. Second,

$$\tan^2\theta = \frac{\frac{1}{3}(Y+a)^2}{(Y+a)^2} = \frac{1}{3}.$$

So $\tan\theta = \frac{1}{\sqrt{3}}$ wherever the point is on the curve: the acute angle between the tangents is constant, and it equals 30°.

Checks

- With $a = 1$, the tangents at $x = 2$ and $x = 0$ are $y = x - 1$ and $y = 0$, at $45°$. They meet at $(1, 0)$, and $(0 + 3)^2 = 9 = 8 + 1^2$.
- For part (iii) with $a = 1$, take $X = 0$: then $(Y + 7)^2 = 48$, so $Y = 4\sqrt{3} - 7 \approx -0.072$. The tangents through $(0, Y)$ touch at $x = \pm\sqrt{-4Y} \approx \pm 0.536$, with gradients ± 0.268. Since $\tan 15° \approx 0.268$, each makes $15°$ with the horizontal and they are $30°$ apart.
- A computer check over a thousand random values of a and points on both branches of each hyperbola confirmed the $45°$ and $30°$ results.

Common slips

- Trying to solve for p and q separately. Only the symmetric combinations $p + q$ and pq are needed, with $(p - q)^2 = (p + q)^2 - 4pq$.
- Forgetting the $\pm$ in part (i): squaring the condition keeps both orientations of the $45°$ angle, and that is what gives a single equation.
- In part (iii), not checking that $Y + a \neq 0$ before cancelling.

Beyond STEP

The tangents from a point (X, Y) to the parabola are governed by the quadratic $t^2 - 2Xt + 4aY = 0$, and the locus of points from which the tangents meet at a fixed angle is always a hyperbola (for $90°$ it degenerates to the directrix $y = -a$). These ideas belong to projective geometry, where the pole–polar relation links a point to its chord of contact. For more questions in this spirit, with commentary on how to attack them, see Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.